

Final Assignment: Reinforcement Learning Applications in Real-World Domains

1. Assignment Overview

In this assignment, each team will explore how Reinforcement Learning can be applied to a real-world application problem.

The goal is **not only to describe RL algorithms**, but to understand how RL problems are formulated and solved in practice.

Each team must choose:

1. **One application field**
2. **One specific application problem**
3. **Three RL algorithms or approaches from the same field**

The selected algorithms must include:

Two Medium-level algorithms + One Advanced-level algorithm

All three algorithms must be applied to the **same application problem**, so that the comparison between them is meaningful.

2. Assignment Goal

By completing this assignment, students should be able to:

- Formulate a real-world problem as an RL problem
 - Identify the agent, environment, state, action, and reward
 - Understand why RL is suitable for a specific application
 - Analyze different RL algorithms in the same problem
 - Compare algorithms based on stability, data needs, action space, and deployment difficulty
 - Evaluate RL systems using appropriate metrics
 - Use AI tools critically as thinking partners, not answer generators
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4. Algorithm Selection Rule

Each team must choose exactly **three algorithms**:

2 Medium-level algorithms + 1 Advanced-level algorithm

All three algorithms must be selected from the **same field** and applied to the **same application problem**.

Example:

Field: Recommendation Systems

Application Problem: Movie Recommendation

Selected Algorithms:

1. Contextual Bandits - Medium
 2. DQN - Medium
 3. Offline RL - Advanced
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5. Possible Fields

Teams may choose one field from the following list:

1. NLP / LLMs
2. Computer Vision
3. Image Generation
4. Optimization
5. Recommendation Systems
6. Robotics
7. Healthcare
8. Finance
9. Autonomous Driving
10. Cloud Resource Management

6. Fields, Algorithms, Levels, and Descriptions

6.1 NLP / LLMs

Possible application problems:

- Chatbot response optimization
- LLM alignment
- Text summarization optimization
- Dialogue policy learning
- Question answering optimization
- Human preference learning
- Prompt optimization

Algorithm / Approach	Level	One-Sentence Description
REINFORCE / Policy Gradient	Medium	Learns to generate responses or text sequences that receive higher reward, such as better quality or stronger user preference.
PPO	Medium	Improves a language model policy while preventing very large policy updates, making it useful for stable LLM fine-tuning.
RLHF	Medium	Uses human feedback to train a reward model and then improves the language model to generate responses preferred by humans.
Actor-Critic	Medium	Uses one model to choose language actions and another model to evaluate how good those actions are.
RLAIF	Advanced	Uses AI-generated feedback instead of human feedback to guide and improve language model behavior.

Algorithm / Approach	Level	One-Sentence Description
DPO	Advanced	Optimizes the model directly from preference pairs without using a traditional RL training loop.
GRPO-style Optimization	Advanced	Optimizes language model behavior using group-based preference or reward signals, often used in advanced reasoning-oriented LLM training.

6.2 Computer Vision

Possible application problems:

- Object localization
- Object detection refinement
- Visual tracking
- Active vision
- Image region selection
- Neural Architecture Search for vision models
- Medical image analysis

Algorithm / Approach	Level	One-Sentence Description
DQN	Medium	Uses a neural network to process visual states and choose actions such as move, zoom, scale, or stop.
Double DQN	Medium	Reduces overestimation of action values in visual decision-making tasks.
Dueling DQN	Medium	Separates the value of the current visual state from the advantage of each possible visual action.
Actor-Critic	Medium	Learns a visual decision policy while also estimating how good each decision is.

Algorithm / Approach	Level	One-Sentence Description
PPO	Medium	Optimizes visual decision-making policies in a stable way, such as active vision or visual navigation.
Attention-based RL	Advanced	Learns where to focus in an image to improve recognition, tracking, or localization.
RL for Neural Architecture Search	Advanced	Uses RL to search for an effective neural network architecture for a computer vision task.
MCTS with Neural Networks	Advanced	Combines search and learned value estimates to select better visual or planning actions.

6.3 Image Generation

Possible application problems:

- Text-to-image generation alignment
- Prompt optimization
- Image editing
- Image enhancement
- Reward-guided image generation
- Human preference optimization for generated images

Algorithm / Approach	Level	One-Sentence Description
PPO	Medium	Updates an image generation policy to produce images with higher reward, such as better prompt alignment or visual quality.
REINFORCE	Medium	Improves image generation using reward feedback from completed generated images.
Preference-based RL	Medium	Learns from comparisons between generated images, such as image A being preferred over image B.

Algorithm / Approach	Level	One-Sentence Description
Bandit-based Prompt Optimization	Medium	Tries different prompts and learns which prompt variations produce better images.
RLHF for Image Generation	Advanced	Uses human feedback on generated images to improve the model toward preferred outputs.
RLAIF for Image Generation	Advanced	Uses AI feedback to score generated images and improve the generation process.
Reward-guided Generation	Advanced	Uses a reward model to guide generation toward goals like aesthetics, safety, or prompt consistency.
DPO-style Optimization	Advanced	Uses preference data to directly improve generated outputs without a full RL loop.

6.4 Optimization

Possible application problems:

- Job scheduling
- Vehicle routing
- Traveling Salesman Problem
- Resource allocation
- Warehouse optimization
- Traffic signal optimization
- Supply chain optimization
- Cloud job scheduling

Algorithm / Approach	Level	One-Sentence Description
DQN	Medium	Uses deep learning to handle large or complex optimization states.
Policy Gradient	Medium	Learns a policy that directly chooses optimization actions to maximize final reward.
REINFORCE	Medium	Uses the final quality of a complete solution to improve future decision sequences.
Actor-Critic	Medium	Combines action selection with value estimation to improve optimization decisions.
PPO	Medium	Provides stable policy optimization for complex decision-making problems.
MCTS	Advanced	Searches through possible future decision paths and selects actions with better estimated outcomes.
Pointer Networks + RL	Advanced	Learns to output an ordered sequence of items, such as cities in TSP or jobs in scheduling.
GNN + RL	Advanced	Represents the optimization problem as a graph and learns decisions over nodes or edges.
Multi-Agent RL	Advanced	Uses multiple agents to solve distributed optimization problems through cooperation or competition.

6.5 Recommendation Systems

Possible application problems:

- Product recommendation
- Movie recommendation
- News recommendation
- Music recommendation

- Video recommendation
- Feed ranking
- Advertisement recommendation
- Personalized learning recommendations

Algorithm / Approach	Level	One-Sentence Description
Contextual Bandits	Medium	Uses the current user context to recommend an item and learn from immediate feedback.
Q-learning	Medium	Learns recommendations that consider how current recommendations affect future user behavior.
DQN	Medium	Uses a neural network to learn recommendation policies from large and complex user states.
Policy Gradient	Medium	Directly optimizes a recommendation policy for long-term user engagement.
PPO	Medium	Learns recommendation policies in a stable way while optimizing long-term satisfaction.
Offline RL	Advanced	Learns from historical user interaction logs without directly experimenting on users.
SlateQ	Advanced	Treats recommendation as choosing a whole list of items instead of a single item.
Conservative Q-Learning	Advanced	Learns from logged recommendation data while avoiding actions that are too different from historical behavior.

6.6 Robotics

Possible application problems:

- Robot arm control
- Object grasping

- Robot navigation
- Drone control
- Humanoid walking
- Path planning
- Collision avoidance
- Sim-to-real transfer

Algorithm / Approach	Level	One-Sentence Description
DQN	Medium	Uses sensory input or images to learn discrete robot control actions.
DDPG	Medium	Learns continuous robot control actions such as joint angles or motor commands.
PPO	Medium	Provides stable policy learning for robot control tasks.
Model-Based RL	Medium	Learns or uses a model of the environment to plan robot actions before executing them.
TD3	Advanced	Improves DDPG by reducing value estimation errors in continuous control.
SAC	Advanced	Learns continuous control policies while encouraging exploration using entropy.
Imitation Learning + RL	Advanced	Starts by imitating expert demonstrations and then improves performance using RL.
Safe RL	Advanced	Trains the robot while avoiding dangerous or unsafe actions.
Hierarchical RL	Advanced	Breaks a complex robot task into smaller skills such as reach, grasp, and lift.

6.7 Healthcare

Possible application problems:

- Treatment recommendation
- ICU decision support
- Drug dosage optimization
- Personalized medicine
- Hospital resource allocation
- Medical decision support

Algorithm / Approach	Level	One-Sentence Description
DQN	Medium	Uses neural networks to learn treatment decisions from complex patient states.
Contextual Bandits	Medium	Selects an intervention based on the current patient state and immediate outcome.
Actor-Critic	Medium	Learns a treatment policy while estimating the expected value of medical decisions.
Model-Based RL	Medium	Models how a patient state may evolve and uses this model for planning.
Offline RL	Advanced	Learns from historical medical records without experimenting directly on patients.
Conservative Q-Learning	Advanced	Avoids actions that are very different from the historical data, which is important in high-risk domains.
Safe RL	Advanced	Optimizes treatment decisions while trying to avoid unsafe actions.
Inverse RL	Advanced	Tries to infer the hidden reward or objective behind expert medical decisions.

Note: This topic must be treated as a research and algorithm analysis task only, not as medical advice.

6.8 Finance

Possible application problems:

- Trading strategy
- Portfolio management
- Asset allocation
- Risk management
- Market making
- Dynamic pricing
- Order execution

Algorithm / Approach	Level	One-Sentence Description
DQN	Medium	Uses neural networks to learn trading or portfolio decisions from complex financial data.
Double DQN	Medium	Reduces overestimation when evaluating financial actions.
Dueling DQN	Medium	Separates the value of the market state from the advantage of each financial action.
Policy Gradient	Medium	Directly optimizes a financial decision policy to improve return or reduce risk.
PPO	Medium	Provides stable training for trading or portfolio allocation policies.
DDPG	Medium	Handles continuous portfolio allocation decisions such as assigning percentages to assets.
SAC	Advanced	Learns continuous financial policies while encouraging exploration.

Algorithm / Approach	Level	One-Sentence Description
Risk-sensitive RL	Advanced	Optimizes decisions while explicitly considering risk, volatility, or drawdown.
Offline RL	Advanced	Learns from historical market data without directly interacting with the market.
Distributional RL	Advanced	Models the distribution of possible returns instead of only the expected return, which is useful for risk-aware financial decisions.

Note: This topic must be treated as an algorithmic research task only, not as financial advice.

6.9 Autonomous Driving

Possible application problems:

- Lane changing
- Adaptive cruise control
- Steering control
- Collision avoidance
- Parking
- Traffic light control
- Autonomous intersection management

Algorithm / Approach	Level	One-Sentence Description
DQN	Medium	Learns discrete driving decisions such as keep lane, change lane, brake, or accelerate.
Double DQN	Medium	Improves the accuracy of action-value estimates for driving decisions.

Algorithm / Approach	Level	One-Sentence Description
Dueling DQN	Medium	Evaluates the driving situation separately from the advantage of each possible action.
PPO	Medium	Learns stable driving policies for tasks such as lane changing or navigation.
DDPG	Medium	Handles continuous driving controls such as steering angle and acceleration.
TD3	Advanced	Improves DDPG by reducing overestimation and instability in continuous driving control.
SAC	Advanced	Learns robust continuous driving policies while encouraging exploration.
Multi-Agent RL	Advanced	Models multiple vehicles as agents that interact with each other on the road.
Safe RL	Advanced	Trains driving policies while avoiding dangerous actions such as collisions.
Model-Based RL	Advanced	Uses a model of vehicle dynamics or traffic behavior to plan actions.
Imitation Learning + RL	Advanced	Starts from human driving demonstrations and then improves using reinforcement learning.

6.10 Cloud Resource Management

Possible application problems:

- Autoscaling
- Job scheduling
- CPU/GPU resource allocation
- Load balancing

- Data center cooling
- Network routing
- Cloud cost optimization
- Queue management

Algorithm / Approach	Level	One-Sentence Description
DQN	Medium	Uses neural networks to make resource decisions from complex system states such as CPU, memory, latency, and queue length.
Double DQN	Medium	Reduces overestimation when choosing resource allocation or scaling actions.
Actor-Critic	Medium	Learns resource management policies while estimating their impact on cost and performance.
PPO	Medium	Provides stable learning for autoscaling, scheduling, or resource allocation policies.
DDPG	Medium	Handles continuous resource allocation decisions such as assigning percentages of CPU, memory, or power.
Contextual Bandits	Medium	Selects a resource configuration based on the current system context and immediate feedback.
SAC	Advanced	Learns continuous resource management policies while encouraging exploration.
Multi-Agent RL	Advanced	Uses multiple agents to manage different servers, services, or workloads.
Offline RL	Advanced	Learns from historical system logs without testing risky decisions directly in production.
Safe RL	Advanced	Avoids resource management decisions that may violate latency, availability, or reliability constraints.

Algorithm / Approach	Level	One-Sentence Description
Model-Based RL	Advanced	Uses a model of system behavior to predict the effect of scaling or scheduling decisions.

7. Required Report Parts

Part A: Problem Formulation

Answer the following questions:

1. What is the selected field and application problem?
2. Why is Reinforcement Learning suitable for this problem?
3. Who or what is the agent?
4. What is the environment?
5. Is the task episodic or continuing?
6. What are the state, action, and reward?

Your answer should clearly explain how the application problem can be formulated as an RL problem.

Part B: Algorithm Analysis

For each of the three selected algorithms, answer the following questions:

1. What does this algorithm do in the selected problem?
2. What is the state?
3. What is the action?
4. What is the reward?
5. Is it value-based, policy-based, actor-critic, bandit, model-based, or offline RL?
6. What is the algorithm flow step by step?

7. What are the advantages?
 8. What are the limitations?
 9. What data, simulator, or feedback is needed?
 10. What can go wrong?
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Part C: Comparison

Compare the three algorithms directly.

Answer the following questions:

1. Compare the three algorithms directly.
 2. Which algorithm is more suitable for discrete actions?
 3. Which algorithm is more suitable for continuous actions?
 4. Which one is more stable during training?
 5. Which one needs more data?
 6. Which one is easier to implement?
 7. Which one is safer for real-world deployment?
 8. Which one would you choose and why?
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Part D: Evaluation

Answer the following questions:

1. What metrics would you use to evaluate the algorithms?
 2. Would you use online RL, offline RL, or simulation?
 3. Why not use only supervised learning?
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Part E: AI Chat Links

Each team must include a table that documents the AI interactions used during the assignment.

Requirements:

- The table must contain exactly 5 rows (one for each team member).
- Each row must include:
 - Student name
 - Selected algorithm
 - Algorithm level (Medium / Advanced)
 - AI chat link used by that student(may be multiple links because of quota)

Algorithm Distribution Rule:

- Each Medium-level algorithm must be assigned to 1 students
- The Advanced-level algorithm must be assigned to 3 students

This means:

- 1 students → Medium Algorithm 1
- 1 students → Medium Algorithm 2
- 3 students → Advanced Algorithm

Important Notes:

- Each student must have their own AI chat interaction (no shared links).
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8. Required AI Usage

8.1 Objective

Students are required to use AI tools such as ChatGPT, Microsoft Copilot, Gemini, Claude, or similar tools as part of the problem-solving process.

However, the goal is **not to get direct answers**.

The goal is to demonstrate:

- Understanding
- Questioning

- Iteration
- Critical thinking
- Error detection
- Improvement of AI-generated ideas

Students should use AI as a **thinking partner**, not as an answer generator.

8.2 Core Rule

The submitted AI interaction must not contain direct answers only.

It must show a real discussion process where the team explores the problem, improves ideas, questions the AI, and reaches a final decision based on reasoning.

A shallow copy-paste answer from AI will lose marks, even if the final answer looks correct.

8.3 Required AI Chat Characteristics

Your AI interaction must clearly demonstrate the following:

1. Problem Exploration

Your chat should show that you explored the problem before jumping to the final answer.

You should ask questions such as:

What does this application problem mean from an RL perspective?

Who is the agent in this problem?

What is the environment?

Can this problem be formulated as an MDP?

Is the problem episodic or continuing?

Why is RL suitable here?

2. Iterative Refinement

Your chat should show that you improved your ideas step by step.

You should try different possible formulations of:

- State
- Action
- Reward
- Agent
- Environment
- Evaluation metrics
- Algorithm choice

Example:

Student: I think the state can be only the user profile.

AI: That could work.

Student: But this ignores the current session behavior. Should we include recent clicks or search history?

AI: Yes, including session behavior can make the state more informative.

Student: Then our improved state will include both long-term user profile and short-term session context.

3. Critical Thinking

Your interaction should show that you questioned the AI response.

You should ask questions such as:

Why is this formulation better?

What if the reward is delayed?

Is this action space discrete or continuous?

Is this algorithm suitable for this type of action?

Could there be a simpler algorithm?

Could this fail in real-world deployment?

Is there a better alternative?

8.4 What Is Not Allowed

You will lose marks if:

- The AI chat contains direct answers only.
 - The interaction is shallow.
 - The chat looks like copy-paste from AI.
 - The same prompt pattern is repeated mechanically.
 - The discussion appears artificial or scripted.
 - You cannot explain your solution during evaluation.
 - You submit AI-generated content without showing your own reasoning.
 - You accept AI suggestions without checking whether they fit your application problem.
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